SST 409: Stochastic Processes

Lecture Notes on Random Variables

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A complete transcription of the 15 handwritten notebook pages. All equations, symbols and notation have been re-typeset in LaTeX.

Abbreviations used in the original: r.v. = random variable; st. pr. = stochastic process; g.f. = generating function; p.g.f. = probability generating function; dist. = distribution; mtd. = method; qty = quantity; prob. = probability.

— p. 1 —

1. Stochastic Processes: Definition and Examples

Here, we mean a set of r.v. $\{X(t)\}$ depending on a real parameter t which varies in a certain set of natural nos.

A st. pr. is denoted by

$$\{X(t),\ t \in T\}.$$

In many problems of theory of st. pr., the parameter t means time. Though it can denote any other physical qty such as space in which the process occurs.

Exp. of st. pr.

1. Random walk where we are concerned with the motion of a particle in a straight line which moves in discrete jumps with certain probabilities i.e. at each stage, it moves one unit to the left or right with probability p or $1-p$. If initially the particle is at the origin then its position $X(n)$ after n jumps is a discrete r.v. depending on the discrete parameter n. Thus $\{X(n),\ n \in T\}$ is a st. pr.

2. The no. of telephone calls $X(t)$ during the arbitrary period of time t is a discrete random variable depending on the continuous parameter t. The family of r.v.'s $\{X(t),\ t \in T\}$ constitutes a st. pr. $\left[T = (0, \infty)\right]$.

Read 3 & 4.

— p. 2 —

2. Generating Functions

Def: Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. Using a variable s we may define a function

$$A(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{k=0}^{\infty} a_k s^k \qquad \cdots(1)$$

If $A(s)$ (power series) converges in some interval $s_0 < s < s_1$, then $A(s)$ is called the g.f. of the sequence $a_0, a_1, a_2, \ldots$. The variable s itself has no particular significance.

Now:

$$\frac{dA(s)}{ds} = a_1 + 2a_2 s + 3a_3 s^2 + 4a_4 s^3 + \cdots + k a_k s^{k-1} + \cdots$$

$$\left.\frac{dA(s)}{ds}\right|_{s=0} = a_1$$

$$\frac{d^2A(s)}{ds^2} = 2a_2 + 3\times 2\,a_3 s + 4\times 3\,a_4 s^2 + \cdots + k(k-1)a_k s^{k-2} + \cdots$$

$$\left.\frac{d^2A(s)}{ds^2}\right|_{s=0} = 2a_2 \quad \Rightarrow \quad a_2 = \frac{1}{2}\left.\frac{d^2A(s)}{ds^2}\right|_{s=0}$$

$$\left.\frac{d^3A(s)}{ds^3}\right|_{s=0} = 3!\,a_3 \quad \Rightarrow \quad a_3 = \frac{1}{3!}\left.\frac{d^3A(s)}{ds^3}\right|_{s=0}$$

$$\left.\frac{d^kA(s)}{ds^k}\right|_{s=0} = k!\,a_k \quad \Rightarrow \quad a_k = \frac{1}{k!}\left.\frac{d^kA(s)}{ds^k}\right|_{s=0}$$

Exp 1

If $a_k = 1$ for all $n \geq 1$, $\{a_k\} = \{1, 1, 1, \ldots\}$, we have $a_0 = 1, a_1 = 1, a_2 = 1, \ldots$

Then the g.f. of $\{a_k\}$ is given by

$$A(s) = \sum_{k=0}^{\infty} a_k s^k = a_0 + a_1 s + a_2 s^2 + a_3 s^3 + \cdots = 1 + s + s^2 + s^3 + \cdots = \frac{1}{1-s}, \qquad |s| < 1$$

Exp 2

Let $\{a_k\} = \left\{\frac{1}{k!}\right\}$.

Then $a_0 = 1, a_1 = 1, a_2 = \frac{1}{2!}, a_3 = \frac{1}{3!}, \ldots$

∴ The g.f. of $\{a_k\}$ is

$$A(s) = \sum_{k=0}^{\infty} a_k s^k = a_0 + a_1 s + a_2 s^2 + a_3 s^3 + \cdots = 1 + s + \frac{s^2}{2!} + \frac{s^3}{3!} + \cdots = e^s$$

Exp 3

Let $\{a_k\} = \left\{\binom{n}{k}\right\}$ for fixed n.

∴ The g.f. of $\{a_k\}$ is

$$A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{n} \binom{n}{k} s^k = (1+s)^n$$

Exp 4

Let $a_0, a_1, a_2, \ldots, a_n$ be a sequence of nos satisfying the recurrence relation

$$a_n + a_{n-1} - 16a_{n-2} + 20a_{n-3} = 0 \qquad \text{for } n \geq 3$$

subjected to the initial values $a_0 = 0, a_1 = 1$ & $a_2 = -1$. Obtain the g.f. of the above sq. $\{a_n\}$.

Soln

Let $A(s) = \sum_{n=0}^{\infty} a_n s^n$ be the g.f. of $\{a_n\}$.

$$\sum_{n=3}^{\infty} a_n s^n + \sum_{n=3}^{\infty} a_{n-1} s^n - 16 \sum_{n=3}^{\infty} a_{n-2} s^n + 20 \sum_{n=3}^{\infty} a_{n-3} s^n = 0$$

$$\sum_{n=3}^{\infty} a_n s^n + s\sum_{n=3}^{\infty} a_{n-1}s^{n-1} - 16s^2\sum_{n=3}^{\infty} a_{n-2}s^{n-2} + 20s^3\sum_{n=3}^{\infty} a_{n-3}s^{n-3} = 0$$

$$A(s) - a_0 - a_1 s - a_2 s^2 + s\big[A(s) - a_0 - a_1 s\big] - 16s^2\big[A(s) - a_0\big] + 20s^3 A(s) = 0$$

— p. 4 —

$$A(s) - 0 - s + s^2 + s\big[A(s) - 0 - s\big] - 16s^2\big[A(s) - 0\big] + 20s^3 A(s) = 0$$

$$A(s)\big[1 + s - 16s^2 + 20s^3\big] = s$$

$$\therefore \quad A(s) = \frac{s}{1 + s - 16s^2 + 20s^3}$$

3. Probability Generating Function (P.G.F.)

Suppose X is a r.v. which assumes non-negative integral values 0, 1, 2, ... and that $\Pr\{X = k\} = p_k, k = 0, 1, 2, \ldots$ and $\sum_k p_k = 1$, then the probability generating function $P(s)$ of the sequence of probabilities $\{p_k\}$ is given by:

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = p_0 + p_1 s + p_2 s^2 + \cdots + p_k s^k + \cdots$$

$$= \sum_{k=0}^{\infty} \Pr\{X = k\}\, s^k = F(s^x)$$

where $F(s^x)$ is the expectation of the function s^x (a r.v.) of the r.v. x.

The series $P(s)$ converges for at least $-1 \le s \le 1$; clearly $P(1) = 1$.

NB: It is possible to have a sequence of probabilities which does NOT form a p.d.f.

e.g. $p_0 = 0, p_1 = \frac{1}{4}, p_2 = \frac{1}{2}, p_3 = \frac{1}{3}, p_4 = \frac{3}{4}$

$$A(s) = p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 = 0 + \frac{1}{4}s + \frac{1}{2}s^2 + \frac{1}{3}s^3 + \frac{3}{4}s^4 = \frac{1}{4}s + \frac{1}{2}s^2 + \frac{1}{3}s^3 + \frac{3}{4}s^4$$

$A(s)$ is a g.f. of $\{p_k\}$ but NOT a p.g.f. since:

$$p_1 + p_2 + p_3 + p_4 \neq 1$$

3.1 Determination of mean & variance using the p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k$$

— p. 5 —

$$\therefore \quad P'(s) = \sum_k k\, p_k s^{k-1}$$

$$P''(s) = \sum_k k(k-1)\, p_k s^{k-2}$$

The expectation $E(x)$ is given by

$$E(x) = \sum_k k\, p_k = P'(1)$$

We have

$$E\{x(x-1)\} = P''(1) = \sum_k k(k-1) p_k$$

and

$$E(x^2) = E\{x(x-1)\} + Ex = P''(1) + P'(1)$$

Hence

$$\text{Var}(x) = E(x^2) - \left[E(x)\right]^2 = P''(1) + P'(1) - \left[P'(1)\right]^2$$

4. The p.g.f. & Moments of Standard Distributions

(i) Bernoulli dist.

Let x have a Bernoulli dist. with parameter p i.e. $\Pr\{X = k\} = p_k = p^k q^{1-k}$, $q = 1 - p, k = 0, 1.$

The p.g.f. of x is given by

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{1} p^k q^{1-k} s^k = q + ps$$

$$\therefore \quad P'(s) = p \qquad E(x) = P'(1) = p$$

$$P''(s) = 0 \quad \Rightarrow \quad P''(1) = 0$$

$$\text{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2 = 0 + p - p^2 = p\left[1 - p\right] = pq$$

(ii) Binomial dist.

Let x have a binomial dist. with parameters n & p i.e.

$$\Pr[X = k] = p_k = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, 2, \ldots, n, \qquad q = 1 - p$$

∴ The p.g.f. of x is given by:

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} s^k = (q + ps)^n$$

$$\therefore \quad P'(s) = n(q + ps)^{n-1} p = np\,(q + ps)^{n-1}$$

$$\therefore \quad E(x) = P'(1) = np\,(q + p)^{n-1} = np \qquad \text{since } q + p = 1$$

$$P''(s) = n(n-1)p^2(q + ps)^{n-2}$$

$$P''(1) = n(n-1)p^2(q + p)^{n-2} = n(n-1)p^2 \qquad \text{since } q + p = 1$$

$$\text{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2$$

$$= n(n-1)p^2 + np - n^2p^2$$

$$= n^2p^2 - np^2 + np - n^2p^2 = np(1 - p) = npq$$

(iii) Geometric dist.

$$\Pr\{X = k\} = p_k = q^k p, \qquad p + q = 1, \qquad k = 0, 1, 2, \ldots$$

The p.g.f. of x is given by:

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} q^k p\, s^k = p \sum_{k=0}^{\infty} (qs)^k$$

$$= p\left(1 + qs + (qs)^2 + (qs)^3 + \cdots\right)$$

$$= p\left[\frac{1}{1-qs}\right] = \frac{p}{1-qs}$$

$$P'(s) = \frac{pq}{(1-qs)^2} \qquad \therefore \quad E(x) = P'(1) = \frac{pq}{(1-q)^2} = \frac{pq}{p^2} = \frac{q}{p}$$

$$P''(s) = \frac{2pq^2}{(1-qs)^3} \qquad P''(1) = \frac{2pq^2}{(1-q)^3} = \frac{2pq^2}{p^3} = \frac{2q^2}{p^2}$$

$$\therefore \quad \text{Var}(x) = P''(1) + P'(1) + \left[P'(1)\right]^2$$

$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$$

$$= \frac{q^2}{p^2} + \frac{q}{p} = \frac{q^2+pq}{p^2} = \frac{q(q+p)}{p^2} = \frac{q}{p^2} \qquad \text{since } q+p=1$$

(iv) Poisson dist. — Read.

— p. 7 —

5. Remarks on the p.g.f.

i) When $s = e^t$, then

$$P\left(e^t\right) = E\left(e^{tx}\right) \qquad \text{which is the Moment generating fn. of } x.$$

ii)

$$\int_0^1 P(s)\,ds = \int_0^1 \left(\sum_{k=0}^{\infty} p_k s^k\right) ds$$

$$= \int_0^1 \left(p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots + p_k s^k\right) ds$$

$$= \int_0^1 p_0\,ds + \int_0^1 p_1 s\,ds + \int_0^1 p_2 s^2\,ds + \int_0^1 p_3 s^3\,ds + \cdots$$

$$= p_0 + \frac{p_1}{2} + \frac{p_2}{3} + \frac{p_3}{4} + \cdots = \sum_{k=0}^{\infty} \frac{p_k}{k+1} = E\left(\frac{1}{X+1}\right)$$

Exp

Let x be a r.v. with p.g.f. $P(s)$. Find the p.g.f. of $Y = mx + n$, where m, n are integers & $m \neq 0$.

Soln

Let $P_Y(s)$ be the p.g.f. of Y. Then;

$$P_Y(s) = E\left(s^Y\right) = E\left[s^{mx+n}\right]$$

$$= E\left(s^{mx}s^n\right) = s^n E\left(s^{mx}\right) = s^n P\left(s^m\right)$$

Remarks

i) We can obtain the probabilities p_k from the p.g.f. p_k can be found from $P(s)$ i.e.

$$p_k = \frac{1}{k!}\left[\frac{d^k P(s)}{ds^k}\bigg|_{s=0}\right]$$

ii) p_k is also given by the co-efficient of s^k in the expansion of $P(s)$ as a power series in s. e.g.

— p. 8 —

i)

$$P(s) = e^{-\lambda(1-s)} = e^{-\lambda}e^{\lambda s}$$

$$= e^{-\lambda}\left[1 + \lambda s + \frac{(\lambda s)^2}{2!} + \frac{(\lambda s)^3}{3!} + \cdots + \frac{(\lambda s)^k}{k!} + \cdots\right]$$

$$\therefore \quad p_k = \text{co-efficient of } s^k = e^{-\lambda}\frac{\lambda^k}{k!} \qquad \text{which is a Poisson Distribution.}$$

6. Sum of Random Variables

a) Fixed numbers

Let X & Y be 2 independent non-negative integral valued random variables. The sum $z = x + y$ is a r.v.

Let $P_z(s)$ be the p.g.f. of Z. Then;

$$P_z(s) = E\left(s^z\right) = E\left(s^{x+y}\right)$$

$$= E\left(s^x\right)E\left(s^y\right) \qquad \text{since } X \text{ \& } Y \text{ are independent}$$

$$= P_x(s)\,P_y(s)$$

[$P_x(s)$: p.g.f. of X; $P_y(s)$: p.g.f. of Y]

This result says that the p.g.f. of the sum of two independent r.v.'s X & Y is the product of the p.g.f. of X & Y.

The result extends to the case when we have n independent random variables $x_1, x_2, x_3, \ldots, x_n$.

That is the p.g.f. of $S_n = x_1 + x_2 + \cdots + x_n$ is the product of the p.g.f.'s of the individual r.v.'s.

$$P_{S_n}(s) = E\left[s^{S_n}\right] = E\left[s^{x_1 + x_2 + \cdots + x_n}\right]$$

$$= E\left(s^{x_1}\right)E\left(s^{x_2}\right)\cdots E\left(s^{x_n}\right) \qquad \text{since } x_1, x_2, \ldots, x_n \text{ are independent}$$

$$= P_{x_1}(s)\,P_{x_2}(s)\cdots P_{x_n}(s)$$

— p. 9 —

If $x_1, x_2, \ldots, x_n$ are also identically distributed, each with p.g.f. $P(s)$, then the p.g.f. of $S_n = x_1 + x_2 + \cdots + x_n$ is

$$\left[P(s)\right]^n$$

Exp

Let x_1 & x_2 be indep. poisson r.v.'s with parameters λ_1 & λ_2 resp. Find the distribution of $Z = X_1 + X_2$.

Soln

Let $P_{x_1}(s)$ & $P_{x_2}(s)$ be the p.g.f.'s of X_1 & X_2 respectively. Then;

$$P_{x_1}(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} e^{-\lambda_1} \frac{\lambda_1^k}{k!} s^k$$
$$= e^{-\lambda_1} \sum_{k=0}^{\infty} \frac{(\lambda_1 s)^k}{k!}$$
$$= e^{-\lambda_1} e^{\lambda_1 s} = e^{-\lambda_1 (1-s)}$$

and similarly; $P_{x_2}(s) = e^{-\lambda_2 (1-s)}$

∴ The p.g.f. of Z is

$$P_z(s) = P_{x_1}(s)\, P_{x_2}(s) = e^{-(\lambda_1 + \lambda_2)(1-s)}$$

This is the p.g.f. of a poisson r.v. with parameters λ_1 & λ_2. This shows the sum of the two indep. poisson r.v.'s is also a poisson r.v.

This result extends to the case of n indep. poisson r.v.'s. That is the sum $S_n = x_1 + x_2 + \cdots + x_n$ of n indep. poisson r.v.'s is a poisson r.v. with mean

$$\sum_{i=1}^{n} \lambda_i$$

where λ_i is the mean of x_i.

b) Random Number

Theorem: Let x_i, $i = 1, 2, \ldots$ be identically & independently dist. (i.i.d.) random variables with $P(x_i = k) = p_k$ & p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k, \qquad i = 1, 2, 3, \ldots$$

Let $S_N = x_1 + x_2 + \cdots + x_N$ where N is a r.v. independent of the x_i's. Let the distribution of N be given by $P[N = n] = g_n$ & p.g.f.

— p. 10 —

of N be

$$G(s) = \sum_{n=0}^{\infty} g_n s^n.$$

Then the p.g.f., $H(s)$, of S_N is given by the compound function $G\big[P(s)\big]$ i.e.

$$H(s) = \sum_{j=0}^{\infty} P(S_N = j)\, s^j = G\big[P(s)\big]$$
$$H'(s) = G'\big[P(s)\big] P'(s)$$
$$\therefore \quad E(S_N) = H'(1) = G'\big[P(1)\big] \cdot P'(1)$$
$$= G'(1)\, P'(1) = E(N)\, E(x_i)$$

Exp

Suppose N has a poisson dist. with mean λ & x_i, $i = 1, 2, \ldots, N$ are i.i.d. with p.g.f. $P(s)$. Find the p.g.f. of $S_N = x_1 + x_2 + \cdots + x_N$.

Soln

p.g.f. of S_N is

$$H(s) = G\big[P(s)\big]$$

But p.g.f. of a poisson r.v. with mean λ is given by;

$$G(s) = e^{-\lambda(1-s)}$$

$$\therefore \quad H(s) = e^{-\lambda\left(1-P(s)\right)}$$

The mean is given by;

$$E(S_N) = E(x_i)\,E(N) = \lambda E(x)$$

7. Tailed or Cumulative Probabilities

i) Upper tail

Let X have a p.d.f. $\Pr\{X = k\} = p_k$, $k = 0, 1, 2, \ldots$ with p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k,$$

and

$$q_k = \Pr\{X > k\} = p_{k+1} + p_{k+2} + \cdots, \qquad k = 0, 1, 2, \ldots$$

with generating funct.

$$\phi(s) = \sum_{k=0}^{\infty} q_k s^k \qquad \left(\sum_k q_k = 1\right)$$

$$\therefore \quad \phi(s) = \sum_{k=0}^{\infty} q_k s^k$$

$$= q_0 + q_1 s + q_2 s^2 + q_3 s^3 + \cdots$$

$$= (p_1 + p_2 + p_3 + \cdots) + (p_2 + p_3 + p_4 + \cdots)s + (p_3 + p_4 + p_5 + \cdots)s^2 + (p_4 + p_5 + p_6 + \cdots)s^3 + \cdots$$

— p. 11 —

$$= p_1 + p_2(1+s) + p_3\left(1 + s + s^2\right) + p_4\left(1 + s + s^2 + s^3\right) + \cdots$$

$$= p_1 + p_2(1+s) + p_3\frac{\left(1-s^3\right)}{(1-s)} + p_4\frac{\left(1-s^4\right)}{(1-s)} + \cdots$$

$$= \frac{p_1(1-s) + p_2\left(1-s^2\right) + p_3\left(1-s^3\right) + p_4\left(1-s^4\right) + \cdots}{(1-s)}$$

$$= \frac{(p_1 + p_2 + p_3 + p_4 + \cdots) - \left(p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right)}{(1-s)}$$

$$= \frac{(p_0 + p_1 + p_2 + p_3 + p_4 + \cdots) - \left(p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right)}{(1-s)}$$

$$\therefore \quad \phi(s) = \frac{1 - P(s)}{1-s}, \qquad -1 < s < 1$$

Alt. mtd.

$$q_k = p_{k+1} + p_{k+2} + p_{k+3} + \cdots$$

$$q_{k-1} = p_k + p_{k+1} + p_{k+2} + p_{k+3} + \cdots$$

$$p_k = q_{k-1} - q_k, \qquad k \geq 1$$

Defining $P(s) = \sum_{k=0}^{\infty} p_k s^k$ & $\phi(s) = \sum_{k=0}^{\infty} q_k s^k$

$$\Rightarrow \quad \sum_{k=1}^{\infty} p_k s^k = \sum_{k=1}^{\infty} q_{k-1} s^k - \sum_{k=1}^{\infty} q_k s^k$$

$$P(s) - p_0 = s \sum_{k=1}^{\infty} q_{k-1} s^{k-1} - \left[\phi(s) - q_0\right]$$

$$P(s) - p_0 = s\,\phi(s) - \phi(s) + q_0 \qquad q_0 = p_1 + p_2 + p_3 + \cdots$$

$$P(s) - (p_0 + q_0) = \phi(s)(s-1)$$

$$\phi(s)(1-s) = (p_0 + q_0) - P(s)$$

$$\phi(s)(1-s) = 1 - P(s) \qquad \text{since } p_0 + q_0 = 1$$

$$\therefore \quad \phi(s) = \frac{1 - P(s)}{1 - s}, \qquad -1 < s < 1$$

— p. 12 —

2) Lower tail

Let $q_k = \Pr\{X \leq k\}, k = 0, 1, 2, \ldots$ & $p_k = \Pr\{X = k\}, k = 0, 1, 2, \ldots$

Defining $P(s) = \sum_{k=0}^{\infty} p_k s^k$ & $\phi(s) = \sum_{k=0}^{\infty} q_k s^k$

$$\therefore \quad q_k = p_k + p_{k-1} + p_{k-2} + \cdots + p_1 + p_0$$

$$q_{k-1} = p_{k-1} + p_{k-2} + \cdots + p_1 + p_0$$

$$\therefore \quad q_k - q_{k-1} = p_k, \qquad k \geq 1 \qquad \Rightarrow (1)$$

Multiplying both sides of the eqn (1) above by s^k & summing over k we obtain;

$$\sum_{k=1}^{\infty} q_k s^k - \sum_{k=1}^{\infty} q_{k-1} s^k = \sum_{k=1}^{\infty} p_k s^k$$

$$\therefore \quad \phi(s) - q_0 - s\,\phi(s) = P(s) - p_0$$

$$\phi(s)(1-s) = P(s) - p_0 + q_0$$

$$\phi(s)(1-s) = P(s) \qquad \text{since } p_0 = q_0$$

$$\therefore \quad \phi(s) = \frac{P(s)}{1 - s}, \qquad -1 < s < 1$$

— p. 13 —

8. Bivariate Probability Generating Functions

Suppose X & Y are integral valued r.v.'s with joint prob. dist.

$$\Pr\{X = j, Y = k\} = p_{jk}, \qquad j = 0, 1, 2, \ldots; \quad k = 0, 1, 2, \ldots$$

and;

$$\sum_j \sum_k p_{jk} = 1$$

Then the joint p.g.f. of X & Y is given by;

$$P(s_1, s_2) = \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} p_{jk}\, s_1^j s_2^k \qquad \text{assuming convergence}$$

for the values of s_1 & s_2 i.e. $|s_1| < 1, |s_2| < 1$.

$$P(s_1, s_2) = E\left(s_1^X s_2^Y\right) \quad \Rightarrow (1)$$

If $s_1 = s_2 = s$, then (1) becomes;

$$P(s, s) = E\left(s^X s^Y\right) = E\left(s^{X+Y}\right)$$

if X & Y are mutually indep. r.v.'s then $P(s_1, s_2)$ becomes;

$$P(s_1, s_2) = E\left(s_1^X\right) E\left(s_2^Y\right)$$

If X & Y are i.i.d., with a common p.g.f. $G(s)$, then we have

$$P(s_1, s_2) = E\left(s_1^X\right) E\left(s_2^Y\right) = G(s_1) \cdot G(s_2) = \left[G(s)\right]^2$$

P.g.f.'s of Marginal Distributions

Let X & Y be 2 r.v.'s, then;

$$P(s_1, s_2) = E\left(s_1^X s_2^Y\right) \quad \rightarrow \text{joint p.g.f.}$$

The p.g.f. of X is given by;

$$P(s_1, 1) = E\left(s_1^X\right)$$

Similarly, p.g.f. of Y is given by;

$$P(1, s_2) = E\left(s_2^Y\right)$$

9. Calculations of Means, Variances & Co-variances of Bivariate Prob. Dist. Functions using the Joint p.g.f.

$$P(s_1, s_2) \quad \rightarrow \quad \text{joint p.g.f. } (x \ \& y)$$
$$P(s_1, 1) = P(s_1) \quad — \quad \text{p.g.f. of } x$$
$$P(1, s_2) = P(s_2) \quad — \quad \text{p.g.f. of } y$$
$$\therefore \quad E(x) = P'(s_1)\big|_{s_1 = 1} = \left.\frac{\partial P(s_1, 1)}{\partial s_1}\right|_{s_1 = 1}$$
$$E(Y) = P'(s_2)\big|_{s_2 = 1} = \left.\frac{\partial P(1, s_2)}{\partial s_2}\right|_{s_2 = 1}$$

The second moment is:

$$E\left[X(X-1)\right] = \left.\frac{\partial^2 P(s_1, 1)}{\partial s_1^2}\right|_{s_1 = 1}$$

and

$$E\left[Y(Y-1)\right] = \left.\frac{\partial^2 P(1, s_2)}{\partial s_2^2}\right|_{s_2 = 1}$$
$$\sigma_x^2 = E\left[X(X-1)\right] + E(x)^2 - \left[E(x)\right]^2$$
$$\sigma_y^2 = E\left[Y(Y-1)\right] + E(Y) - \left[E(Y)\right]^2$$
$$\text{cov}(x, y) = \sigma_{xy} = E(xy) - E(x)E(y)$$

but;

$$E(xy) = \left.\frac{\partial^2 P(s_1, s_2)}{\partial s_1 \partial s_2}\right|_{s_1 = s_2 = 1}$$

Exp

Consider a series of Bernoulli trials with prob. of success p. Suppose that X denotes the no. of failures preceeding the 1st success & Y the no. of failures following 1st success & preceeding 2nd success. Then the joint prob. distribution of X & Y is given by;

$$p_{jk} = \Pr\{X = j, Y = k\} = q^{j+k}p^2; \qquad j,k = 0,1,2; \qquad p + q = 1$$

— p. 15 —

Soln

The joint p.g.f. is

$$P(s_1, s_2) = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} p_{jk}\, s_1^j s_2^k$$

$$= \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} q^{j+k}p^2\, s_1^j s_2^k$$

$$= p^2 \sum_{j=0}^{\infty} (qs_1)^j \sum_{k=0}^{\infty} (qs_2)^k$$

$$= p^2\left[\left(1 + qs_1 + (qs_1)^2 + \cdots\right)\left(1 + qs_2 + (qs_2)^2 + \cdots\right)\right]$$

$$P(s_1, s_2) = p^2\left(\frac{1}{1 - qs_1}\right)\left(\frac{1}{1 - qs_2}\right) = \frac{p^2}{(1 - qs_1)(1 - qs_2)}$$

$$P(s_1) = P(s_1, 1) = \frac{p^2}{(1 - qs_1)(1 - q)} = \frac{p}{1 - qs_1}, \qquad \text{since } q + p = 1$$

$$\therefore \quad P'(s_1) = \frac{pq}{(1 - qs_1)^2}$$

$$\therefore \quad E(x) = P'(1) = \frac{pq}{(1 - q)^2} = \frac{pq}{p^2} = \frac{q}{p}$$

$$P''(s_1) = \frac{2pq^2}{(1 - qs_1)^3}$$

$$P''(1) = \frac{2pq^2}{(1 - q)^3} = \frac{2pq^2}{p^3} = \frac{2q^2}{p^2}$$

$$\therefore \quad \text{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2$$

$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2} = \frac{q^2}{p^2} + \frac{q}{p}$$

$$= \frac{q^2 + pq}{p^2} = \frac{q(q + p)}{p^2} = \frac{q}{p^2} \qquad \text{since } q + p = 1$$

Similarly;

$$E(y) = \frac{q}{p} \qquad \& \qquad \text{Var}(y) = \frac{q}{p^2}$$

$$\frac{\partial P(s_1, s_2)}{\partial s_1} = \frac{p^2 q}{(1 - qs_1)^2(1 - qs_2)}$$

$$\therefore \quad \frac{\partial^2 P(s_1, s_2)}{\partial s_1 \partial s_2} = \frac{p^2 q^2}{(1 - qs_1)^2(1 - qs_2)^2}$$

Appendix A. Notes on the Original

The following are minor notational slips in the handwritten original. They have been transcribed above exactly as written; the corrected forms are given here for study purposes.

1. **p. 6 (Geometric distribution, variance).** The original writes $\mathrm{Var}(x) = P''(1) + P'(1) + \left[P'(1)\right]^2$; the sign of the last term should be minus: $\mathrm{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2$. The subsequent line, $= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$, is handled correctly and the final answer $\mathrm{Var}(x) = q/p^2$ is right.
2. **p. 14 (Bivariate variances).** The original writes $\sigma_x^2 = E[X(X-1)] + E(x)^2 - \left[E(x)\right]^2$; the first moment term should be $E(x)$, i.e. $\sigma_x^2 = E[X(X-1)] + E(x) - \left[E(x)\right]^2$. (Compare p. 5, where $E(x^2) = E\{x(x-1)\} + Ex$ is used correctly.) The companion formula for σ_y^2 is written correctly.
3. **p. 8, item (i).** This is the continuation of the Poisson p.g.f. started on p. 7, item (ii) of "Remarks"; the notes read out p_k as the coefficient of s^k in $P(s) = e^{-\lambda(1-s)}$, giving the Poisson probabilities $p_k = e^{-\lambda}\lambda^k/k!$.
4. **p. 14, example.** The original gives $j, k = 0, 1, 2, \ldots$ (the "…" is implied by the geometric-type structure); strictly, X and Y range over all non-negative integers.

Appendix B. Page Index (source images)

Page	Source image	Content
1	`bg1.png`	Definition of a stochastic process; two examples (random walk, telephone calls)
2	`bg2.png`	Generating functions: definition, coefficient extraction by differentiation, Exp 1
3	`bg3.png`	Exp 2 ($1/k!$), Exp 3 ($\binom{n}{k}$), Exp 4 (recurrence relation, set-up)
4	`bg4.png`	Completion of Exp 4; probability generating function; NB; mean & variance set-up
5	`bg5.png`	Mean/variance from the p.g.f.; Bernoulli and Binomial distributions
6	`bg6.png`	Binomial (contd.), Geometric distribution; Poisson (to be read)

Page	Source image	Content
7	`bg7.png`	Remarks: mgf, integral of $P(s)$, p.g.f. of $Y = mx + n$, recovering p_k
8	`bg8.png`	Poisson p.g.f. expanded; sum of independent random variables (fixed number)
9	`bg9.png`	i.i.d. case; sum of two Poisson variables; statement of the random-sum theorem
10	`bga.png`	Compound (random-sum) p.g.f. $G[P(s)]$; mean; Poisson N; upper tail
11	`bgb.png`	Upper tail continued (two methods)
12	`bgc.png`	Lower tail
13	`bgd.png`	Bivariate p.g.f.; independence; marginal p.g.f.'s
14	`bge.png`	Means, variances, covariance from the joint p.g.f.; example set-up
15	`bgf.png`	Example solved: joint p.g.f., marginals, $E(x)$, $\mathrm{Var}(x)$, cross-derivative